

House Price Predictor

Linear Regression on California Housing Dataset

Artificial Intelligence & Machine Learning — Task 1

Submitted by:	Internship Candidate
Organization:	Maincrafts Technology
Date:	June 21, 2026
Tools Used:	Python, Jupyter Notebook, scikit-learn, pandas, matplotlib, seaborn
Dataset:	California Housing Dataset (scikit-learn built-in)

1. Introduction

This report presents the development and evaluation of a House Price Predictor using Linear Regression, built as part of the Artificial Intelligence and Machine Learning internship program at Maincrafts Technology. The primary objective of this task is to demonstrate the end-to-end Machine Learning workflow — from data loading and exploratory analysis to model training, evaluation, and interpretation of results.

The California Housing Dataset, a well-known benchmark dataset built into scikit-learn, was used for this project. It contains 20,640 records of housing block data from the 1990 California census, with 8 numerical features and a continuous target variable representing the median house value.

Features in the Dataset:

MedInc	Median income of households in the block
HouseAge	Median age of houses in the block
AveRooms	Average number of rooms per household
AveBedrms	Average number of bedrooms per household
Population	Total population of the block
AveOccup	Average number of occupants per household
Latitude	Geographic latitude of the block
Longitude	Geographic longitude of the block
MedHouseVal	TARGET — Median house value (in \$100,000s)

2. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the structure, distribution, and relationships within the dataset before building the model. Key findings are summarized below.

2.1 Dataset Overview

Metric	Value
Total Records	20,640
Total Features	8 (+ 1 target)
Missing Values	0 (dataset is clean)
Target Variable	MedHouseVal (Median House Value)
Target Range	0.15 to 5.0 (in \$100,000s)

2.2 Key EDA Findings

- No missing values were found — the dataset is clean and required no imputation.
- The target variable (MedHouseVal) is right-skewed, with a notable spike at 5.0 due to price capping in the original census data.
- Median Income (MedInc) has the strongest positive correlation with house prices (correlation ~ 0.69), making it the most influential feature.
- House Age shows a weak positive correlation with price, suggesting older homes in California tend to be in desirable locations.
- Latitude and Longitude show moderate negative correlation, reflecting geographic pricing trends (coastal areas command higher prices).
- Average Rooms and Average Bedrooms have moderate positive correlations with price.

3. Model Building

3.1 Data Preprocessing

The dataset was split into training and testing sets using an 80/20 ratio with a fixed random state of 42 to ensure reproducibility. All 8 features were used as inputs to the model.

Split	Records	Percentage
Training Set	16,512	80%
Testing Set	4,128	20%
Total	20,640	100%

3.2 Algorithm — Linear Regression

Linear Regression is a supervised machine learning algorithm that models the relationship between a dependent variable (house price) and one or more independent variables (features) by fitting a linear equation to the observed data. The model learns the optimal coefficients (weights) for each feature during training by minimizing the sum of squared residuals.

The equation learned by the model takes the form: $\text{Price} = w_1 \cdot \text{MedInc} + w_2 \cdot \text{HouseAge} + w_3 \cdot \text{AveRooms} + \dots + \text{bias}$

4. Model Evaluation & Results

4.1 Performance Metrics

The trained model was evaluated on the unseen test set (4,128 records) using three standard regression metrics:

Metric	Value	Interpretation
MAE (Mean Absolute Error)	0.5332	On average, predictions are off by ~\$53,320
RMSE (Root Mean Squared Error)	0.7456	Larger errors are penalized more heavily
R ² (R-Squared Score)	0.5758	Model explains 57.58% of price variance

4.2 Interpretation of Results

The model achieved an R² score of 0.5758, meaning it explains approximately 57.6% of the variance in house prices. While this is a reasonable baseline for a simple linear regression model, it also indicates that the relationship between the features and house prices is not entirely linear.

The MAE of 0.5332 (in units of \$100,000) translates to an average prediction error of \$53,320 per house. Given the complexity of real estate pricing, this is an expected result for a first-iteration linear model without feature engineering or scaling.

4.3 Visualizations

- Price Distribution Histogram — Showed the target variable is right-skewed with a hard cap at 5.0.
- Correlation Heatmap — Confirmed MedInc as the strongest predictor of house prices.
- Feature vs Price Scatter Plots — Visualized individual relationships between each feature and the target.
- Actual vs Predicted Plot — Points close to the diagonal red line indicate accurate predictions.
- Residual Plot — Residuals spread around zero, suggesting the model has no systematic bias, though variance increases at higher price points.

5. Improvement Ideas

The following techniques could significantly improve model performance beyond the current baseline:

Technique	Benefit
Feature Scaling (StandardScaler)	Normalizing features to the same scale can improve model convergence and performance.
Ridge / Lasso Regression	Regularized regression techniques reduce overfitting and handle multicollinearity better.
Polynomial Features	Adding squared or interaction terms can capture non-linear relationships in the data.
Random Forest Regressor	An ensemble tree-based model that typically achieves R^2 of 0.80+ on this dataset.
Gradient Boosting (XGBoost)	State-of-the-art boosting algorithm that can achieve even higher accuracy.
Outlier Removal	Removing or capping the price-capped houses (at 5.0) could improve model learning.
Cross-Validation	Using k-fold cross-validation gives a more robust estimate of model performance.

6. Conclusion

This project successfully demonstrated the complete Machine Learning workflow — from data loading and exploratory analysis to model training, evaluation, and result interpretation. A Linear Regression model was trained on the California Housing Dataset, achieving an R^2 score of 0.5758, an MAE of 0.5332, and an RMSE of 0.7456 on the test set.

The project provided hands-on experience with Python's core data science libraries including pandas, NumPy, scikit-learn, matplotlib, and seaborn. The trained model was saved as a pickle file for future deployment. The results establish a solid baseline, and several improvement strategies have been identified for future iterations of the model.